VINCENT BONNIN

Robotics Engineering Student | Mechanical Design, Embedded Electronics & Control
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PROFESSIONAL SUMMARY

Engineering student at École Centrale de Lille working across the full robot development cycle: mechanical design and 3D modelling, embedded electronics, and real-time control (Python, C++, ROS2). Currently completing a gap-year internship in embedded control at Compliance Robotics. Seeking a 4 to 6 month robotics internship abroad from April 2027.

EDUCATION

École Centrale de Lille | Lille, France | **2024 - Present**

MEng, General Engineering - robotics and mechatronics track

- Top-tier French "Grande École" of engineering; coursework in mechanics, automatic control, embedded systems, signal processing and computer science.

Lycée Joffre - CPGE PC* (Classes Préparatoires aux Grandes Écoles) | Montpellier, France | **2021 - 2024**

Intensive undergraduate programme in mathematics and physics

- Two-year selective programme (approx. 16 h/week of advanced mathematics and physics) preparing the nationwide competitive entrance exams to French engineering schools; PC* denotes the highest-level track.
- Baccalauréat, Science stream, Highest Honours (Mention Très Bien), 2021.

PROFESSIONAL EXPERIENCE

Robotics Engineering Intern - Compliance Robotics | Lille, France | **Sep 2026 - Mar 2027**

- Design full embedded electronics architecture for a soft robotic manipulator, including sensor and actuator interfacing.
- Develop control firmware and deploy on target hardware.
- Optimise sensor placement and electronics integration to improve system performance.

Mechatronics Research Intern - Inria | Montpellier, France | **Feb - Mar 2025**

- Designed, 3D-modelled and assembled a vibration generator used to calibrate mechanomyography sensors.
- Implemented closed-loop control of the vibration output to meet calibration accuracy requirements.

ROBOTICS PROJECTS

Autonomous Mobile Robot & Robotic Arm - ROS2 Simulation & Deployment | **June 2026**

- Modelled, simulated and validated autonomous navigation in Gazebo (obstacle avoidance, real-time sensing), then ported control code to physical robot (2-wheel drive, LiDAR); tuned sensor thresholds and motor commands for real-world performance.
- Designed a 4-DOF robotic arm in Onshape, imported to ROS2 workspace and deployed inverse kinematics controller for end-effector positioning.

Autonomous Robot - French Robotics Cup 2026 | La Roche-sur-Yon, France | **May 2026**

- Designed and built a miniature differential-drive robot (1.4 kg, 20×20 cm, ESP32-based) with a suction-cup actuator within a 9-person student team; personally led 3D CAD modelling, mechanical prototyping and component integration.

Alstom Robotics Battle Hackathon | Saint-Ouen, France | **May 2026**

- Developed the control interface in Python for a colour-sensing differential-drive robot in a 5-person team over a one-day hackathon on railway use cases.

Mechanical Hand Prosthesis - École Centrale de Lille | **May - Jul 2025**

- Delivered a fully mechanical hand prosthesis for a finger-amputation user as part of a 4-person project team; led the sizing, 3D modelling and 3D-printed prototype.

TECHNICAL SKILLS

Programming & Frameworks: Python, C++, ROS2
CAD: SolidWorks, CATIA, Onshape, Meshmixer
Simulation & Computation: MATLAB, COMSOL, Gazebo
Business Tools: Salesforce, Looker Studio, Excel
Languages: French (native), English (C1), Mandarin Chinese (basic, HSK 2)

ACTIVITIES

Amateur competitive boxing (licensed). Co-led business development for student engineering consultancy (€48k revenue, 15 members). Trumpet: Conservatory CEM diploma; winner of the L'Isle-sur-la-Sorgue trumpet competition, 2016 and 2018.